

1 Overview

In this paper, Ruetsche contextualizes and criticizes the position of Effective Realism in the scientific realism debate, ultimately arguing that RG methods/EFTs better support a “humble empiricism.” The basic structure of the debate is reduced to:

- scientific realists, faced with the PMI, opt for Psillos’ *divide et impera* strategy: identify a selection principle to choose what aspects of some theory T are worthy of belief. However, unless this principle is precisely specified (what persists/“structure” tend to be inadequate or begging the question), opponents tend to be unconvinced.
- empiricists argue instead that success of T warrants belief not in T’s truth but in its empirical adequacy. Opponents argue this is a “selective skepticism” about observables/unobservables, and requires one to believe in the persistence of regularities without an underlying reason to expect them (agnosticism about hidden springs)

Ruetsche points out that both positions face issues therefore with circumscription: realists need to circumscribe defensible knowledge claims, empiricists needs to circumscribe a sustainable skepticism.

Effective Realism is supposed to offer a new position in this debate because it argues that all interacting QFTs are merely effective, and need not be true at all scales. In the relevant regimes, they approximate the fundamental physics. EFT realists use RG methods as their specific, and physically-motivated selection principle. Ruetsche will argue that resisting empiricism will commit Effective Realists to a kind of *fundamentalism*.

2 Motivating Effective Realism

Free QFTs (i.e. non-interacting) are mathematically well-understood, and can be show to satisfy certain axiomatic characterizations. Interacting QFTs are not as well-understood – we don’t know how to solve their equations of motion, nor do axiomatic approaches help us understand them¹. Instead, we solve these systems via perturbation theory, in which solutions are found as approximations to the free theory.

$$\begin{aligned}\mathcal{L}_0 &= \frac{1}{2}\partial^2\phi^2 - \frac{1}{2}m^2\phi^2 \\ \mathcal{L}_I &= -\lambda\frac{\phi^4}{4!}\end{aligned}\tag{1}$$

When λ is small, the full Lagrangian $\mathcal{L} \equiv \mathcal{L}_0 + \mathcal{L}_I$ can be seen as a small disturbance to the free and well-understood Lagrangian $\mathcal{L}$. Particles are given by modes (or excitations) of the free field, so particles of $\mathcal{L}$ are given by excitations of mass m weakly interacting with themselves. When λ is small, we see interacting solutions may be expressed as:

$$\phi(x) = \phi_0(x) + \lambda\phi_1(x) + \lambda^2\phi_2(x) + \dots\tag{2}$$

If λ is small, then we can use a few of these terms (rather than the whole expansion) to calculate physically interesting quantities like: n-point functions $\rightarrow$ S-matrix elements $\rightarrow$ experimental cross-sections². We cannot calculate n-point functions for interacting QFTs, so instead we approximate them perturbatively. Terms in this expansion are represented diagrammatically as Feynman Diagrams.

It turns out that individual terms in this series actually diverge.³

$$\int_0^\infty \frac{k^3}{k^2 + m^2} dk \mapsto \int_0^\Lambda \frac{k^3}{k^2 + m^2} dk\tag{3}$$

To tame these divergences we 1) regularize the integral (i.e. introduce a high-momentum cutoff)⁴ and 2) renormalize (relieve the cut-off dependence of the regularized theory by re-writing the regularized

¹actually, this needs a qualification! Perturbative AQFT can offer some insight

²need each to calculate the next

³QFT is famous for having divergences of different kinds. These are UV divergences (aka from high momentum contributions) but there’s also IR (low p) divergences, surface divergences, and the divergence of the whole series

⁴or change the number of dimensions or introduce fictitious particles

Lagrangian in terms of new coefficients and new fields, rather than the original “bare” quantities, so that as $\Lambda \rightarrow \infty$ the individual terms converge at each order, and the values of these renormalized masses/amplitudes/couplings are determined experimentally.

$$\begin{aligned} \mathcal{L}_B(\phi_B, m_B, \lambda_B) &\mapsto \mathcal{L}_R(\phi_R, m_R, \lambda_R) \\ m_R &= Z_m m_B \end{aligned} \tag{4}$$

If we can tame the UV divergences of a theory by reparameterizing a finite set of coefficients appearing in the Lagrangian, we call it *perturbatively renormalizable*. James Fraser has pointed out that the real problem here is: why do these approximations work as well as they do? Renormalization does not tell us anything about what is being approximated. Perturbative QFT on its own offers no clues about how to construct a physical picture that would explain its success.⁵

2.1 RG resolution

RG methods are meant to answer the real problem: instead of extracting an explanation for the success of an interacting QFT from itself, we argue that interacting QFTs are merely effective.

T is **merely effective** just in case T, while itself not a complete and accurate account of physical reality, approximates that account whatever it is within a restricted domain of application.

Ruetsche offers some concerns that this does not solve the “real problem”:

1. This resolution does not engage with any particulars of pQFT
2. RG analysis on its own does not tell us what the final theory is, and if our favorite pQFTs are the effective theories induced by some $\mathcal{L}_{final}$
3. It is a live possibility that $\mathcal{L}^l$ will depend on infinitely many fields, which will not be perturbatively renormalizable⁶

RG methods are supposed to resolve issue (1). Briefly, consider $\mathcal{L}$ from before. We can re-interpret any particular theory as a list of coefficients for the coupling coefficients of products of fields and their derivatives. For ϕ^4 -theory, most of these coefficients are 0. We define a Lagrangian $\mathcal{L}$ which governs phenomena at experimentally accessible scales l , where $\mathcal{L}^l \in \mathcal{T}$. We can then relate Lagrangians at different scales, say $\Lambda > l$ by a map $R_{l\Lambda} : \mathcal{T} \rightarrow \mathcal{T}$ via:

$$\mathcal{L}^l = R_{l\Lambda} \mathcal{L}^\Lambda \tag{5}$$

We call $\mathcal{L}^l$ the effective Lagrangian which approximates $\mathcal{L}^\Lambda$ at scale l . $R_{l\Lambda}$ is part of a family of transformations on $\mathcal{T}$ forming a semi-group. It’s not a true group because we can’t invert this map after coarse-graining (i.e. we cannot extract the high energy physics from the low-energy physics after it’s been coarse-grained).

This leads us to issues (2) and (3). Ruetsche then offers 2 desiderata for resolving these concerns. The action of RG on $\mathcal{T}$ must be s.t.:

1. All high energy theories flow to the same subspace $\mathcal{T}_{eff}^l$ of $\mathcal{T}$ – resolves (2)
2. The surface of attraction $\mathcal{T}_{eff}^l$ is finite-dimensional (we only need finitely-many parameters to specify a Lagrangian in $\mathcal{T}_{eff}^l$) – resolves (3)

If these conditions are fulfilled, we’ll call these theories *effectively specifiable*. In these cases, the contribution of unknown physics to physics at experimentally-accessible scales can be experimentally determined.

(Q) There’s no guarantee we live in an effectively-specifiable world like this, why require it?

⁵this is actually an active area of research, Fraser’s more recent work concerns extracting information about the perturbative series from the success of approximations

⁶is this a real concern?

- Polchinski has shown that the space of theories whose Lagrangian are polynomials in 4-dim scalars and their derivatives – both free and disturbed by a wide range of weakly-coupled interactions – agree in their empirical predictions of $\mathcal{L}^l$ and $\mathcal{L}^\Lambda$ (i.e. determine the same n-point functions at scale l), has a 3-dim surface of attraction, and share the form of the perturbative renormalizable ϕ^4 -theory we got by UV-divergence taming
- No similarly rigorous demonstrations exist for interacting QFTs BUT looking to QCD, the effective Lagrangian at low-energy atomic scales exhibits quark confinement. ⁷

RG solution to the real problem: perturbatively-renormalizable Lagrangians correspond to finitely-specifiable effective theories. They encapsulate the low-energy implications of unknown higher-energy physics, which is tractable because $\mathcal{T}_{eff}^l$ is finite dimensional. Even if our renormalization schemes are suspect and opportunistic, we'll get the physics right.

2.2 Effective Realism

Attributed to Williams, Effective Realism is an attitude toward the empirically-successful effective theories in light of RG methods. In response to critiques of ill-defined selection principles, the Effective Realist declares a successful theory T is *effective* and therefore:

- actually incorporates domain restrictions into itself
- restricts its content in the sense that RG methods tell us what elements of a given EFT are invariant across different ways of modeling high-energy physics that is approximated by the same low-energy physics (i.e. common elements on the surface of attraction). These are the stable and robust features we invest our realism in via “interpretation.”

example: mirror fermions in lattice QCD depend on how the theory is latticized in the high-E/small- a regime (i.e. choice of spacing a). This fails the robustness criteria, so we need not be realists about that aspect of the effective theory.

According to Williams, Effective Realism is meant to be a foil to the so-called “Standard Interpretation”: an account of theory T 's truth conditions, where we consider the description of the fundamental ontological structure of the world offered by T 's truth conditions to be perfectly accurate and exhaustive. If we insist on a Standard Interpretation, then we suppress a theory's capacity to explain physics at different scales, and will be misled about genuine ontological information by interpreting a theory only at the scales at which it becomes unreliable and conflate that with structure at all scales.

3 Perturbing Realism

3.1 How resistant to the PMI is ER?

(ER response to PMI) no matter what the future of physics holds, specific features of our best current physics will persist in that future physics. RG analysis helps us identify those features, and gives us reason to expect their persistence.

(Critique) not quite: the most RG methods can tell us is, *only if* $T_{final} \in \mathcal{T}$, features robust under RG flows will persist in future physics⁸. Our best current theories may be dramatically-incompetent guides to how the physical world is

consider Newtonian Gravity, since LUG + perturbative corrections can save the phenomena, we may endorse the common features shared across these corrections – i.e. central gravitational forces acting instantaneously on flat space vs. in GR, we dispense with these central forces and attribute phenomena to curved space (NG gets the ontology wrong)

⁷unclear to me how this tells me that QCD really is effectively-specifiable

⁸(Q) future physics $\neq$ physics at higher scales

3.2 How realist is ER?

(ER realism) we are realists about robust features, i.e. features invariant under RG flow (i.e. the finite couplings parameterizing $\mathcal{T}_{eff}^l$, the n-point functions defined by a Lagrangian in $\mathcal{T}_{eff}^l$, the particles correlate to the fields appearing in effective Lagrangians of $\mathcal{T}_{eff}^l$).

(Critique: Couplings) the quantities corresponding to the couplings m_R, λ_R, ϕ_R are always in $\mathcal{T}_{eff}^l$ even though their values change. Empiricists would be happy to endorse these couplings as phenomena without being realists⁹

(Critique: Correlation Functions) if the ER is distinctly realist, the challenge is to interpret these n-point functions in a realist way. IF these were free QFTs, we could use the Wightman reconstruction theorem¹⁰ to justify commitment to these as commitment to the underlying theory. But interacting QFTs do not satisfy the WRT, so the ER needs to *interpret* these correlation functions ontologically, rather than just phenomena/book-keeping devices for S-matrix elements – i.e. interpret $\langle \phi(x)\phi^\dagger(y) \rangle$ as: a photon of momentum k is created at x while a photon of momentum k is annihilated at y .¹¹

ERs need more than just the robustness criteria to distinguish themselves from the empiricists, they need interpretation.

(Critique: Fields) Ruetsche argues the empiricist can also commit to the existence of particles as the entities corresponding to fields in $\mathcal{T}_{eff}^l$, going beyond that requires more interpretation

(Q) what does “commitment” mean for Ruetsche’s empiricist?

Does the ER need to disavow interpretation wholesale? For a successful theory T , we can ask:

1. What does T say the world is like?
2. Why does T work as well as it does?

(1) is a question about how to interpret T , which enables understanding of T . (2) asks what explains T ’s success, which may or may NOT involve a claim that T is true.

(Fundamentalism) A proper understanding of why T succeeds requires understanding T , because T succeeds as well as it does because the world is the way T says it is. No exogenous/empirical features of T are required to explain T ’s success, it is explanatorily self-sufficient. Therefore, T ’s domain must be unrestricted, and its descriptions of that domain totally accurate.

The generalist corollary holds because otherwise, something outside of T is required to explain why T is successful in some regimes but not others, or why things aren’t exactly as T says they are.

Ruetsche argues that interpretation on its own does not commit one to fundamentalism: we can understand T (explicate its truth conditions) without believing in T (can explicate (1) without using its answer for (2))¹². EFTs avoid fundamentalism since something else empirical (?) and exogenous to T (namely T_{final}) are needed to explain T ’s success.

“Fundamentalist’s sin isn’t to interpret T . It’s to decide that, having interpreted T , no further explanatory work remains – and thereby suppress the possibility that T is merely effective.”

⁹very confused about what “endorsement” means for the empiricists

¹⁰a QFT satisfying the Wightman axioms can be reconstructed from/is equivalent to a full set of n-point functions

¹¹this is shaky ground, correlation functions are terms in a perturbative expansion of the time-ordered product of operator-valued distributions

¹²don’t we still need some sort of bridge principle? Otherwise this is a miracle. Consider useful but avowedly wrong models, like the liquid drop model of the atom – no one thinks that atom really is a liquid, but modeling it as such simplifies a lot of calculations and offers new insights into nuclear fission. But I would still consider it an important question to answer why it works as well as it does, that relates its features to features of the underlying physics. An answer to the question would probably be most satisfactory if it says something like: the liquid drop model of the atom is a useful effective model because it has the same dependency relations between its model objects that really is going on inside the atom. This is more than just: it gets it right because it gets the empirical predictions right, but is less than, it gets it right because it has the right ontology.

Ruetsche argues that this highlights that ER actually supports a variety of empiricism: we can construct explanations of T's success which makes no appeal to T's truth.

(*Humble Empiricism*) In experimentally-accessible regimes, T approximates T_{final} 's predictions (not T_{final} 's account of physical reality)¹³

The humble empiricist likes answers to question (2) that are mediated not by an interpretation of the theory nor a commitment to its truth, but instead that T succeeds because it approximates some underlying true theory in its domain of application.

(Q) how is this an answer? this seems like a truism, given by the fact that T is successful?

Differences from other kinds of empiricism:

- not as radical as Cartwright-ian non-realists who don't believe in the coherence of some T_{final}
- still offers an explanation, just not a fundamentalist one¹⁴

HE allows for agnosticism about the hidden springs while justifying faith in continued empirical success because many distinct underlying theories flow to $\mathcal{T}_{eff}^l$ (T_{final} is underdetermined)

4 Effective Empiricism

(Criterion) The particles corresponding to fields ψ_i^{eff} exist

The effective empiricist (EE) will *pragmatically endorse* the Criterion, in a way that avoids 3 puzzles that the ER must face:

1. (mirror fermions) the ER seems to be committed to criterion, since even though the details of their appearance depend on lattice spacing, the fact of their appearance does not. For the EE, we are only committed to mirror fermions as useful for organizing phenomena at some scale. Since we do not observe mirror fermions, we need not be committed to them as empiricists¹⁵.
2. (relevance) physicists distinguish between coefficients in $\mathcal{T}_{eff}^l$ that are *relevant* at scale l , and *negligible* at scale l (aka suppressed by factors of l/Λ_{UV}). Which ones should we take (criterion) to endorse? For the ER, this is a puzzle since we usually consider existence/non-existence to be sharp. For the EE, since pragmatic commitment comes in degrees, we have no puzzle.
3. (scale relativity) the commitments of (criterion) are scale-relative – not an issue for EE since utility is indexed to us and our experimental ability.

Final example: quasi-particles in solid-state physics.

- ‘dressed’ electron: ‘bare’ electrons traveling through semi-conductors accrete a cloud of positive charge, treated as one unit in the effective theory
- phonon: collective excitations of the protons/neutrons/electrons in a lattice whose energy is quantized, which behave like sound waves.

Physicists distinguish between these two: dressed electrons are supposedly better candidates for particles than phonons because there are particles present, just not the ones in our effective physics. This presents an issue for (Criterion) for both parties that Ruetsche argues the EE is better able to handle than the ER, since it complicates the ‘real but not fundamental’ category. The ER cannot make sense of the distinction made here, since the distinction depends on the underlying T' that induces T_{eff} which the ER regards as irrelevant to the robustness criteria.

(Q) should we try to recover this distinction at all?

¹³Ruetsche sees this position as stable even if we do not think a T_{final} exists; now instead we just need an underlying theory T' for any successful theory T

¹⁴I'm not sure I buy this

¹⁵again, committed in what way?

References